Example 1 - rivt Doc

Example 1 - rivt Doc

Attn: User Example

proj. 0001

Table of Contents

0.1 | Summary

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam using E-B theory [1] . It also serves as an annotated example of a single rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the **<** rivt user manual **>** .

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command at the end of each rivt file (Doc-API rv.D). Published files are found in the _published folder.

0.1 - 2 | Load Combinations

Dead and live loads effects are taken from ASCE 7-05 [2]

Table 1: Load Effects (stored: t001-1.csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

0.1 - 3 | Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the rvsrc folder. The description is the table title, followed by the max column width.

Table 3: Beam Geometry (rvsrc/beam1.csv)

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

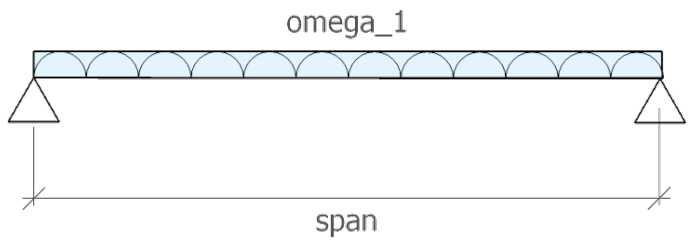

Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: Dead load [ASCE7-05 2.3.2]

$$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$$

dl_1 = 3.64 klf [dl_1] = 53.09 kN_m | Dead load [ASCE7-05 2.3.2]

D_3	spc_1	D_4	D_2	D_1
10.00 psf	2.00 ft	3.00 klf	2.10 psf	3.80 psf
partitions DL	beam spacing	fixed machinery DL	plywood DL	joists DL

Eq. 2: Live load [ASCE7-05 2.3.2]

$$ll_1 = 1.6 \cdot L_1 \cdot spc_1$$

ll_1 = 0.13 klf [ll_1] = 1.87 kN_m | Live load [ASCE7-05 2.3.2]

spc_1	L_1
2.00 ft	40.00 psf

```
beam spacing  ASCE7-05 LL
────────────  ───────────
```

Eq. 3: Total load [ASCE7-05 2.3.2]

```
ω₁ = dl₁ + ll₁

ω₁ = 3.77 klf     [ω₁] = 54.96 kN_m   | Total load [ASCE7-05 2.3.2]

dl₁                          ll₁
───────────────────────────  ───────────────────────────
3.64 klf                     128.00 ft·psf
─────                        ─────
Dead load [ASCE7-05 2.3.2]   Live load [ASCE7-05 2.3.2]
───────────────────────────  ───────────────────────────
```

0.1 - 4 | Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions (rvsrc/sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S (sectprop.py)

```
section₁ = rectsect(b₁, h₁)

section₁ = 540.00 in3     [section₁] = 8849.01 cm3   | rectangle - S (sectprop.py)

b₁          h₁
──────────  ──────────
10.00 inch  18.00 inch
─────       ─────
beam width  beam depth
──────────  ──────────
```

Eq. 5: rectangle - I (sectprop.py)

```
inertia₁ = rectinertia(b₁, h₁)

inertia₁ = 4860.0 in4     [inertia₁] = 202288.5 cm4    | rectangle - I (sectprop.py)

b₁          h₁
──────────  ──────────
10.0 inch   18.0 inch
─────       ─────
beam width  beam depth
──────────  ──────────
```

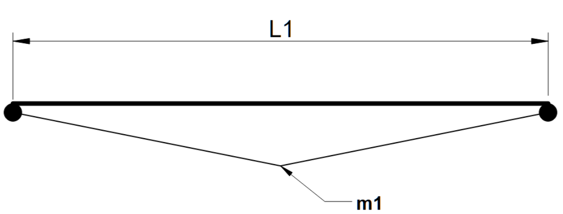

Fig. 2 - Moment diagram

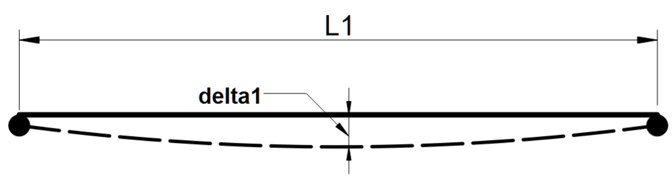

Fig. 3 - Deflection diagram

Maximum bending stress formula

Eq.6:

```
     M₁
σ₁ = ──
     S₁
```

Eq. 7: Mid-span UDL moment

```
             2
     ω₁·spn₁
m₁ = ────────
        8

m₁ = 120.52 ftkip     [m₁] = 163.40 mkN    | Mid-span UDL moment

spn₁       ω₁
─────────  ───────────────────────────
16.00 ft   3.77 klf
```

beam span	Total load [ASCE7-05 2.3.2]

Eq. 8: Bending stress

$$fb_1 = \frac{m_1}{section_1}$$

fb_1 = 2678.2 lb_in2 [fb_1] = 18.5 MPA | Bending stress

$section_1$	m_1
540.0 inch3	120.5 ft2·klf
rectangle - S (sectprop.py)	Mid-span UDL moment

Eq.9: Stress ratio

$fb_1 < Fb_1$

[1] fb_1	[2] Fb_1	ratio [1]/[2]	check	reference
2.68 ksi	20.00 ksi	0.13	OK	Stress ratio

Eq. 10: mid-span deflection (sectprop.py)

δ_1 = midspan_δ(spn_1, ω_1, E_1, $inertia_1$)

δ_1 = 0.04 inch [δ_1] = 1.00 mm | mid-span deflection (sectprop.py)

spn_1	E_1	ω_1	$inertia_1$
16.00 ft	29000.00 ksi	3.77 klf	4860.00 inch4
beam span	modulus of elasticity	Total load [ASCE7-05 2.3.2]	rectangle - I (sectprop.py)

1 “Euler–Bernoulli beam theory,” Wikipedia, Wikimedia Foundation. [Online].https://en.wikipedia.org/wiki/Euler_Bernoulli_beam_theory.[Accessed: Jun. 15, 2026].

2 ASCE/SEI 7-05, Minimum Design Loads for Buildings and Other Structures,American Society of Civil Engineers, 2005.